

Plant and animal protein intake and its association with overweight and obesity among the Belgian population.



The objective of the present study was to assess animal and plant protein intakes in the Belgian population and to examine their relationship with overweight and obesity (OB). The subjects participated in the Belgian National Food Consumption Survey conducted in 2004. Food consumption was assessed by using two non-consecutive 24 h dietary recalls. About 3083 participants (≥ 15 years of age; 1546 males, 1537 females) provided completed dietary information. Animal protein intake (47 g/d) contributed more to total protein intakes of 72 g/d than plant protein intake, which accounted for 25 g/d. Meat and meat products were the main contributors to total animal protein intakes (53 %), whereas cereals and cereal products contributed most to plant protein intake (54 %). Males had higher animal and plant protein intakes than females ($P < 0.001$). Legume and soya protein intakes were low in the whole population (0.101 and 0.174 g/d, respectively). In males, animal protein intake was positively associated with BMI ($\beta = 0.013$; $P = 0.001$) and waist circumference (WC; $\beta = 0.041$; $P = 0.002$). Both in males and females, plant protein intake was inversely associated with BMI (males: $\beta = -0.036$; $P < 0.001$; females: $\beta = -0.046$; $P = 0.001$) and WC (male: $\beta = -0.137$; $P < 0.001$; female: $\beta = -0.096$; $P = 0.024$). In conclusion, plant protein intakes were lower than animal protein intakes among a representative sample of the Belgian population and decreased with age. Associations with anthropometric data indicated that plant proteins could offer a protective effect in the prevention of overweight and OB in the Belgian population.